



Nice Netcat...

Nice netcat... 



Easy

General Skills

picoCTF 2021

AUTHOR: SYREAL

Description

There is a nice program that you can talk to by using this command in a shell:

Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.

Its current status is:

NOT_RUNNING

Launch Instance

Hints 

1 2

151,098 users solved



92%

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 picoCTF{FLAG}

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Challenge: Nice netcat

Category: General Skills

Date: 01/11/26



I. Objective

The objective of this task was to identify the hidden flag by connecting to a remote server using the netcat (**nc**) utility. Instead of analyzing or downloading a file, the challenge required establishing a network connection and correctly interpreting the data sent by the server.

II. Background

In networking-based capture-the-flag (CTF) challenges, participants often interact with remote servers rather than local files. A remote server is simply another computer on the internet that waits for connections from users. These servers listen on specific ports, which act like numbered doors that allow different services to run simultaneously. By connecting to the correct port using the appropriate tool, information such as messages or flags can be transmitted to the user. This challenge introduces basic networking concepts and demonstrates how data can be shared over a network connection.

III. Tool Used

- **Oracle VirtualBox (Kali Linux)** – used as the Linux environment required to run the challenge
- **Linux Terminal** – used to execute networking commands
- **nc (netcat)** – a networking utility used to connect to a remote server and receive data
- **Hostname** – the address of the remote server provided by the challenge
- **Port** – the specific service endpoint on the remote server used for the connection

IV. Methodology

1. Kali Linux was launched using Oracle VirtualBox.
2. The hostname and port number provided by the challenge were noted.



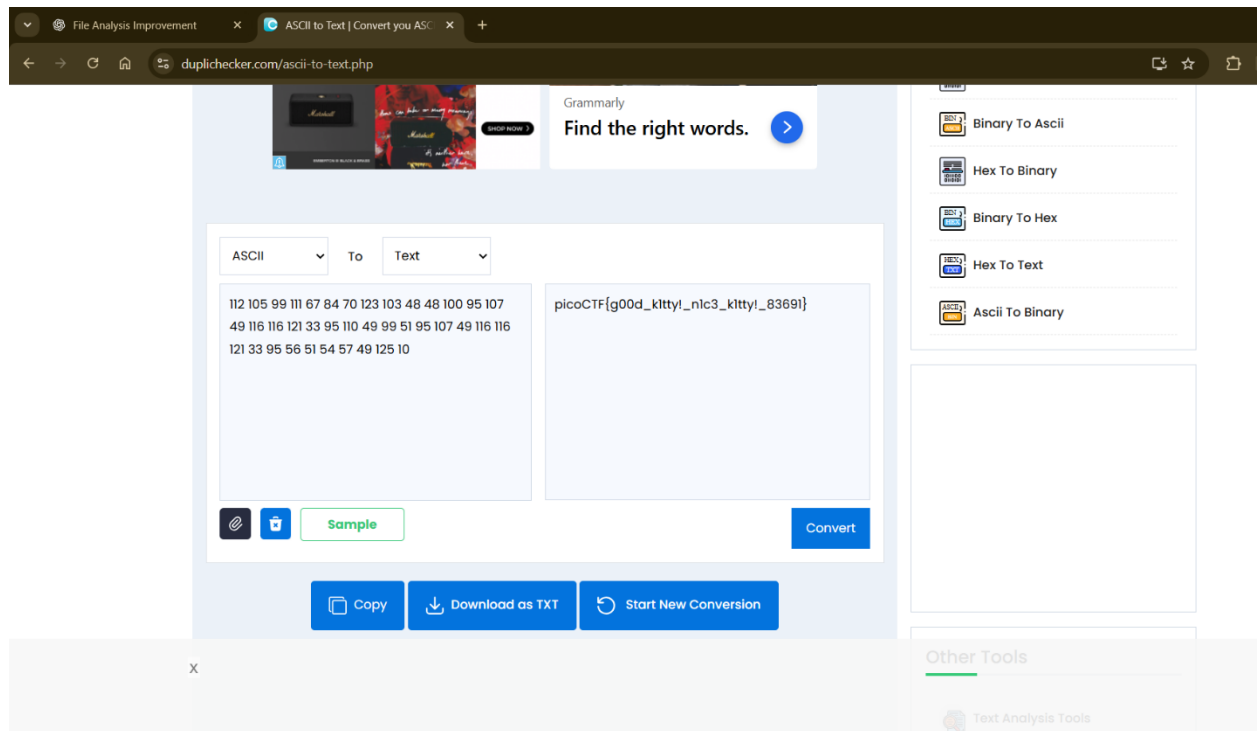
3. A connection to the remote server was established using the command `nc <hostname> <port>`.
4. Upon connecting, a vertical list of numbers was displayed in the terminal.

```
(kali㉿kali)-[/home/kali]
PS> nc wily-courier.picocftf.net 63628
112
105
99
111
67
84
70
123
103
48
48
100
95
107
49
116
116
121
33
95
110
49
99
51
95
107
49
116
116
121
33
95
56
51
54
57
49
125
10
```

5. Based on the challenge hints and the nature of the output, the numbers were identified as decimal ASCII values rather than readable English text.
6. The ASCII values were copied and converted into readable text using an online ASCII-to-text converter.
7. After conversion, the decoded text revealed the flag.

V. Result

picoCTF{g00d_k1tty!_n1c3_k1tty!_83691}



VI. Explanation

Unlike previous challenges such as “Obedient Cat” and “Wave a Flag,” this task did not involve downloading or opening a file. Instead, the challenge relied entirely on communicating with a remote server. A remote server can be thought of as another computer that is always ready to talk to users who connect to it. When a connection is made using netcat, the server sends data directly to the user’s terminal.

In this challenge, the server transmitted the flag as decimal ASCII values, one number per line. Netcat displayed the raw data exactly as it was received, without converting it into readable text. By recognizing the output as ASCII and converting it into characters, the hidden flag became readable.

Whoever you are, it is to note that the port or maybe even the hostname provided would be different from yours, so do not copy the port and the hostname unless it’s the same.

VII. Conclusion

This challenge emphasized the importance of understanding basic networking concepts such as remote servers, ports, and raw data transmission. It demonstrated how netcat can be used to interact with network services and retrieve information directly



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from a server. Additionally, the task reinforced the value of recognizing common data encodings like ASCII and knowing how to convert them into readable form. These fundamental skills are essential for both CTF challenges and real-world cybersecurity scenarios.

— **The Analyst: Hyposelenia**